THE GENERAL RESOLUTION LEMMA FOR GEOMETRICALLY REGULAR TARGETS IN POSITIVE CHARACTERISTIC

THE STACKS PROJECT AUTHORS

Editorial note. The source marks one step '(minor detail omitted)' in Ad (item-choose-deltai), namely the clearing-denominators choice of s realizing both displayed conditions; it is the mechanical denominator step of the construction and is carried as an explicit core obligation.

Editorial note. Inputs lemma-improve-presentation (07CE), lemma-compare-standard (07EZ) and lemma-solution-modulo (07FH) are published entries used as standing inputs by their display numbers; not proven on the author's behalf.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3182 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Presentation improvement with retraction. lemma-improve-presentation (same chapter, PUBLISHED 07CE). Standardizes the auxiliary algebra B and gives the section C -> B.
- Comparison of standardness conditions. lemma-compare-standard (same chapter, PUBLISHED 07EZ). Promotes smoothness of C_{x_i} with free differentials to strict standardness of x_i^c.
- Artinian approximation factorization. lemma-solution-modulo (same chapter, PUBLISHED 07FH). Supplies the step (item-NP) factorization k[y]_p/p^n -> D -> Lambda_q/q^n.
- Ogoma's annihilator lemma. lemma-ogoma (carded L26814). Used in the steps (item-modify-pii) and (item-choose-deltai) to enforce the annihilator equalities.
- Enlarging the Artinian approximation ring. lemma-enlarge-solution-modulo (carded L26817). Supplies the multipliers delta_i of step (item-choose-deltai) with the (a)(b)(c) properties.
- Removing strictly standard elements from a resolution. lemma-lift-solution (carded L26812). Justifies the reduction (item-second-resolve) modulo the powers gamma_i^e.
- Height-zero delocalization. lemma-delocalize-height-zero (carded L26813). Justifies the reduction (item-third-resolve) to the localization at r.
- Strictly standard elements survive base change. lemma-strictly-standard-base-change (carded L26796). Carries strict standardness of gamma_i^c from x_i^c to the base-changed algebra.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 00ON: focused-seeds/geometry/S9-smo-resolve-general/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05CK: focused-seeds/geometry/S9-smo-resolve-general/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00NQ: focused-seeds/geometry/S9-smo-resolve-general/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07DW: focused-seeds/geometry/S9-smo-resolve-general/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07DV: focused-seeds/geometry/S9-smo-resolve-general/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07EZ: sources/stacks-smoothing/pinned-original-source.tex, source lines 936–936. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FA: sources/stacks-smoothing/pinned-original-source.tex, source lines 2447–2447. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FI: sources/stacks-smoothing/pinned-original-source.tex, source lines 2841–2841. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CE: sources/stacks-smoothing/pinned-original-source.tex, source lines 559–559. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07F8: sources/stacks-smoothing/pinned-original-source.tex, source lines 2314–2314. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FC: sources/stacks-smoothing/pinned-original-source.tex, source lines 2504–2504. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FH: sources/stacks-smoothing/pinned-original-source.tex, source lines 2769–2769. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CC: sources/stacks-smoothing/pinned-original-source.tex, source lines 512–512. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07F7: sources/stacks-smoothing/pinned-original-source.tex, source lines 2296–2296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $k \to A \to \Lambda \supset \mathfrak{q}$ be as in Situation 07F7 where*

1. *k is a field of characteristic $p > 0$,*
2. *Λ is Noetherian and geometrically regular over k,*
3. *$\mathfrak{q}$ is minimal over $\mathfrak{h}_A$.*

Then $k \to A \to \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. The lemma is proven by the following steps in the given order. We will justify each of these steps below.

1. Pick an integer $N > 0$ such that $\mathfrak{q}^N \Lambda_\mathfrak{q} \subset H_{A/k}\Lambda_\mathfrak{q}$.
2. Pick generators $a_1, \ldots, a_t \in A$ of the ideal $H_{A/R}$.
3. Set $d = \dim(\Lambda_\mathfrak{q})$.
4. Set $B = A[x_1, \ldots, x_d, z_{ij}]/(x_i^{2N} - \sum z_{ij}a_j)$.
5. Consider B as a $k[x_1, \ldots, x_d]$-algebra and let $B \to C$ be as in Lemma 07CE. We also obtain a section $C \to B$.
6. Choose $c > 0$ such that each x_i^c is strictly standard in C over $k[x_1, \ldots, x_d]$.
7. Set $n = N + dc$ and $e = 8c$.
8. Let $E \subset \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$ be the images of generators of A as a k-algebra.
9. Choose an integer m and a k-algebra map $\varphi : k[y_1, \ldots, y_m] \to \Lambda$ and a factorization by local Artinian rings

$$k[y_1, \ldots, y_m]_\mathfrak{p}/\mathfrak{p}^n k[y_1, \ldots, y_m]_\mathfrak{p} \to D \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$$

such that the first arrow is essentially smooth, the second is flat, E is contained in D, with $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$ the map $k[y_1, \ldots, y_m]_\mathfrak{p} \to \Lambda_\mathfrak{q}$ is flat, and $\mathfrak{p}\Lambda_\mathfrak{q} = \mathfrak{q}\Lambda_\mathfrak{q}$.

(10) Choose $\pi_1, \ldots, \pi_d \in \mathfrak{p}$ which map to a regular system of parameters of $k[y_1, \ldots, y_m]_\mathfrak{p}$.

(11) Let $R = k[y_1, \ldots, y_m, t_1, \ldots, t_m]$ and $\gamma_i = \pi_i t_i$.

(12) If necessary modify the choice of π_i such that for $i = 1, \ldots, d$ we have

$$\mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)R}(\gamma_i) = \mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

(13) There exist $\delta_1, \ldots, \delta_d \in \Lambda$, $\delta_i \notin \mathfrak{q}$ and a factorization $D \to D' \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$ with D' local Artinian, $D \to D'$ essentially smooth, the map $D' \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$ flat such that, with $\pi_i' = \delta_i \pi_i$, we have for $i = 1, \ldots, d$
   (a) $(\pi_i')^{2N} = \sum a_j \lambda_{ij}$ in Λ where $\lambda_{ij} \bmod \mathfrak{q}^n\Lambda_\mathfrak{q}$ is an element of D',
   (b) $\mathrm{Ann}_{\Lambda/(\pi'^e_1, \ldots, \pi'^e_{i-1})}(\pi'_i) = \mathrm{Ann}_{\Lambda/(\pi'^e_1, \ldots, \pi'^e_{i-1})}(\pi'^2_i)$,
   (c) $\delta_i \bmod \mathfrak{q}^n\Lambda_\mathfrak{q}$ is an element of D'.

(14) Define $B \to \Lambda$ by sending x_i to π_i' and z_{ij} to λ_{ij} found above. Define $C \to \Lambda$ by composing the map $B \to \Lambda$ with the retraction $C \to B$.

(15) Map $R \to \Lambda$ by φ on $k[y_1, \ldots, y_m]$ and by sending t_i to δ_i. Further introduce a map

$$k[x_1, \ldots, x_d] \longrightarrow R = k[y_1, \ldots, y_m, t_1, \ldots, t_d]$$

by sending x_i to $\gamma_i = \pi_i t_i$.

(16) It suffices to resolve

$$R \to C \otimes_{k[x_1, \ldots, x_d]} R \to \Lambda \supset \mathfrak{q}$$

(17) Set $I = (\gamma_1^e, \ldots, \gamma_d^e) \subset R$.

(18) It suffices to resolve

$$R/I \to C \otimes_{k[x_1, \ldots, x_d]} R/I \to \Lambda/I\Lambda \supset \mathfrak{q}/I\Lambda$$

(19) We denote $\mathfrak{r} \subset R = k[y_1, \ldots, y_m, t_1, \ldots, t_d]$ the inverse image of $\mathfrak{q}$.

(20) It suffices to resolve

$$(R/I)_\mathfrak{r} \to C \otimes_{k[x_1, \ldots, x_d]} (R/I)_\mathfrak{r} \to \Lambda_\mathfrak{q}/I\Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}/I\Lambda_\mathfrak{q}$$

(21) Set $J = (\pi_1^e, \ldots, \pi_d^e)$ in $k[y_1, \ldots, y_m]$.

(22) It suffices to resolve

$$(R/JR)_\mathfrak{p} \to C \otimes_{k[x_1, \ldots, x_d]} (R/JR)_\mathfrak{p} \to \Lambda_\mathfrak{q}/J\Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}/J\Lambda_\mathfrak{q}$$

(23) It suffices to resolve

$$(R/\mathfrak{p}^nR)_\mathfrak{p} \to C \otimes_{k[x_1, \ldots, x_d]} (R/\mathfrak{p}^nR)_\mathfrak{p} \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$$

(24) It suffices to resolve

$$(R/\mathfrak{p}^nR)_\mathfrak{p} \to B \otimes_{k[x_1, \ldots, x_d]} (R/\mathfrak{p}^nR)_\mathfrak{p} \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$$

(25) The ring $D'[t_1, \ldots, t_d]$ is given the structure of an $R_\mathfrak{p}/\mathfrak{p}^nR_\mathfrak{p}$-algebra by the given map $k[y_1, \ldots, y_m]_\mathfrak{p}/\mathfrak{p}^nk[y_1, \ldots, y_m]_\mathfrak{p} \to D'$ and by sending t_i to t_i. It suffices to find a factorization

$$B \otimes_{k[x_1, \ldots, x_d]} (R/\mathfrak{p}^nR)_\mathfrak{p} \to D'[t_1, \ldots, t_d] \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$$

where the second arrow sends t_i to δ_i and induces the given homomorphism $D' \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$.

(26) Such a factorization exists by our choice of D' above.

We now give the justification for each of the steps, except that we skip justifying the steps which just introduce notation.

Ad (1). This is possible as $\mathfrak{q}$ is minimal over $\mathfrak{h}_A = \sqrt{H_{A/k}\Lambda}$.

Ad (6). Note that A_{a_i} is smooth over k. Hence B_{a_j}, which is isomorphic to a polynomial algebra over $A_{a_j}[x_1, \ldots, x_d]$, is smooth over $k[x_1, \ldots, x_d]$. Thus B_{x_i} is smooth over $k[x_1, \ldots, x_d]$. By Lemma 07CE we see that C_{x_i} is smooth over $k[x_1, \ldots, x_d]$ with finite free module of differentials. Hence some power of x_i is strictly standard in C over $k[x_1, \ldots, x_n]$ by Lemma 07EZ.

Ad (9). This follows by applying Lemma 07FH.

Ad (10). Since $k[y_1, \ldots, y_m]_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ is flat and $\mathfrak{p}\Lambda_{\mathfrak{q}} = \mathfrak{q}\Lambda_{\mathfrak{q}}$ by construction we see that $\dim(k[y_1, \ldots, y_m]_{\mathfrak{p}}) = d$ by Algebra, Lemma 00ON. Thus we can find $\pi_1, \ldots, \pi_d \in \Lambda$ which map to a regular system of parameters in $\Lambda_{\mathfrak{q}}$.

Ad (12). By Algebra, Lemma 00NQ any permutation of the sequence $\pi_1, \ldots, \pi_d$ is a regular sequence in $k[y_1, \ldots, y_m]_{\mathfrak{p}}$. Hence $\gamma_1 = \pi_1 t_1, \ldots, \gamma_d = \pi_d t_d$ is a regular sequence in $R_{\mathfrak{p}} = k[y_1, \ldots, y_m]_{\mathfrak{p}}[t_1, \ldots, t_d]$, see Algebra, Lemma 07DW. Let $S = k[y_1, \ldots, y_m] \setminus \mathfrak{p}$ so that $R_{\mathfrak{p}} = S^{-1}R$. Note that $\pi_1, \ldots, \pi_d$ and $\gamma_1, \ldots, \gamma_d$ remain regular sequences if we multiply our π_i by elements of S. Suppose that

$$\mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)R}(\gamma_i) = \mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

holds for $i = 1, \ldots, t$ for some $t \in \{0, \ldots, d\}$. Note that $\gamma_1^e, \ldots, \gamma_t^e, \gamma_{t+1}$ is a regular sequence in $S^{-1}R$ by Algebra, Lemma 07DV. Hence we see that

$$\mathrm{Ann}_{S^{-1}R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)}(\gamma_i) = \mathrm{Ann}_{S^{-1}R/(\gamma_1^e, \ldots, \gamma_{i-1}^e)}(\gamma_i^2).$$

Thus we get

$$\mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_t^e)R}(\gamma_{t+1}) = \mathrm{Ann}_{R/(\gamma_1^e, \ldots, \gamma_t^e)R}(\gamma_{t+1}^2)$$

after replacing π_{t+1} by $s\pi_{t+1}$ for some $s \in S$ by Lemma 07FC. By induction on t this produces the desired sequence.

Ad (13). Let $S = \Lambda \setminus \mathfrak{q}$ so that $\Lambda_{\mathfrak{q}} = S^{-1}\Lambda$. Set $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$. Suppose that we have a $t \in \{0, \ldots, d\}$ and $\delta_1, \ldots, \delta_t \in S$ and a factorization $D \to D' \to \bar{\Lambda}$ as in (13) such that (a), (b), (c) hold for $i = 1, \ldots, t$. We have $\pi_{t+1}^N \in H_{A/k}\Lambda_{\mathfrak{q}}$ as $\mathfrak{q}^N\Lambda_{\mathfrak{q}} \subset H_{A/k}\Lambda_{\mathfrak{q}}$ by (1). Hence $\pi_{t+1}^N \in H_{A/k}\bar{\Lambda}$. Hence $\pi_{t+1}^N \in H_{A/k}D'$ as $D' \to \bar{\Lambda}$ is faithfully flat, see Algebra, Lemma 05CK. Recall that $H_{A/k} = (a_1, \ldots, a_t)$. Say $\pi_{t+1}^N = \sum a_j d_j$ in D' and choose $c_j \in \Lambda_{\mathfrak{q}}$ lifting $d_j \in D'$. Then $\pi_{t+1}^N = \sum c_j a_j + \epsilon$ with $\epsilon \in \mathfrak{q}^n\Lambda_{\mathfrak{q}} \subset \mathfrak{q}^{n-N}H_{A/k}\Lambda_{\mathfrak{q}}$. Write $\epsilon = \sum a_j c_j'$ for some $c_j' \in \mathfrak{q}^{n-N}\Lambda_{\mathfrak{q}}$. Hence $\pi_{t+1}^{2N} = \sum(\pi_{t+1}^N c_j + \pi_{t+1}^N c_j')a_j$. Note that $\pi_{t+1}^N c_j'$ maps to zero in $\bar{\Lambda}$; this trivial but key observation will ensure later that (a) holds. Now we choose $s \in S$ such that there exist $\mu_{t+1j} \in \Lambda$ such that on the one hand $\pi_{t+1}^N c_j + \pi_{t+1}^N c_j' = \mu_{t+1j}/s^{2N}$ in $S^{-1}\Lambda$ and on the other $(s\pi_{t+1})^{2N} = \sum \mu_{t+1j}a_j$ in Λ (minor detail omitted). We may further replace s by a power and enlarge D' such that s maps to an element of D'. With these choices μ_{t+1j} maps to $s^{2N}d_j$ which is an element of D'. Note that $\pi_1, \ldots, \pi_d$ are a regular sequence of parameters in $S^{-1}\Lambda$ by our choice of φ. Hence $\pi_1, \ldots, \pi_d$ forms a regular sequence in $\Lambda_{\mathfrak{q}}$ by Algebra, Lemma 00NQ. It follows that ${\pi'}_1^e, \ldots, {\pi'}_t^e, s\pi_{t+1}$ is a regular sequence in $S^{-1}\Lambda$ by Algebra, Lemma 07DV. Thus we get

$$\mathrm{Ann}_{S^{-1}\Lambda/({\pi'}_1^e, \ldots, {\pi'}_t^e)}(s\pi_{t+1}) = \mathrm{Ann}_{S^{-1}\Lambda/({\pi'}_1^e, \ldots, {\pi'}_t^e)}((s\pi_{t+1})^2).$$

Hence we may apply Lemma 07FC to find an $s' \in S$ such that

$$\mathrm{Ann}_{\Lambda/(\pi'^e_1,\ldots,\pi'^e_t)}((s')^q s\pi_{t+1}) = \mathrm{Ann}_{\Lambda/(\pi'^e_1,\ldots,\pi'^e_t)}(((s')^q s\pi_{t+1})^2).$$

for any $q > 0$. By Lemma 07FI we can choose q and enlarge D' such that $(s')^q$ maps to an element of D'. Setting $\delta_{t+1} = (s')^q s$ and we conclude that (a), (b), (c) hold for $i = 1, \ldots, t+1$. For (a) note that $\lambda_{t+1j} = (s')^{2Nq}\mu_{t+1j}$ works. By induction on t we win.

Ad (16). By construction the radical of $H_{(C\otimes_{k[x_1,\ldots,x_d]}R)/R}\Lambda$ contains $\mathfrak{h}_A$. Namely, the elements $a_j \in H_{A/k}$ map to elements of $H_{B/k[x_1,\ldots,x_n]}$, hence map to elements of $H_{C/k[x_1,\ldots,x_n]}$, hence $a_j \otimes 1$ map to elements of $H_{C\otimes_{k[x_1,\ldots,x_d]}R/R}$. Moreover, if we have a solution $C \otimes_{k[x_1,\ldots,x_n]} R \to T \to \Lambda$ of

$$R \to C \otimes_{k[x_1,\ldots,x_d]} R \to \Lambda \supset \mathfrak{q}$$

then $H_{T/R} \subset H_{T/k}$ as R is smooth over k. Hence T will also be a solution for the original situation $k \to A \to \Lambda \supset \mathfrak{q}$.

Ad (18). Follows on applying Lemma 07F8 to $R \to C \otimes_{k[x_1,\ldots,x_d]} R \to \Lambda \supset \mathfrak{q}$ and the sequence of elements $\gamma_1^c, \ldots, \gamma_d^c$. We note that since x_i^c are strictly standard in C over $k[x_1, \ldots, x_d]$ the elements γ_i^c are strictly standard in $C \otimes_{k[x_1,\ldots,x_d]} R$ over R by Lemma 07CC. The other assumption of Lemma 07F8 holds by steps (12) and (13).

Ad (20). Apply Lemma 07FA to the situation in (18). In the rest of the arguments the target ring is local Artinian, hence we are looking for a factorization by a smooth algebra T over the source ring.

Ad (22). Suppose that $C \otimes_{k[x_1,\ldots,x_d]} (R/JR)_\mathfrak{p} \to T \to \Lambda_\mathfrak{q}/J\Lambda_\mathfrak{q}$ is a solution to

$$(R/JR)_\mathfrak{p} \to C \otimes_{k[x_1,\ldots,x_d]} (R/JR)_\mathfrak{p} \to \Lambda_\mathfrak{q}/J\Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}/J\Lambda_\mathfrak{q}$$

Then $C \otimes_{k[x_1,\ldots,x_d]} (R/I)_\mathfrak{r} \to T_\mathfrak{r} \to \Lambda_\mathfrak{q}/I\Lambda_\mathfrak{q}$ is a solution to the situation in (20).

Ad (23). Our $n = N + dc$ is large enough so that $\mathfrak{p}^n k[y_1, \ldots, y_m]_\mathfrak{p} \subset J_\mathfrak{p}$ and $\mathfrak{q}^n\Lambda_\mathfrak{q} \subset J\Lambda_\mathfrak{q}$. Hence if we have a solution $C \otimes_{k[x_1,\ldots,x_d]} (R/\mathfrak{p}^n R)_\mathfrak{p} \to T \to \Lambda_\mathfrak{q}/\mathfrak{q}^n\Lambda_\mathfrak{q}$ of (22 then we can take T/JT as the solution for (23).

Ad (24). This is true because we have a section $C \to B$ in the category of R-algebras.

Ad (25). This is true because D' is essentially smooth over the local Artinian ring $k[y_1, \ldots, y_m]_\mathfrak{p}/\mathfrak{p}^n k[y_1, \ldots, y_m]_\mathfrak{p}$ and

$$R_\mathfrak{p}/\mathfrak{p}^n R_\mathfrak{p} = k[y_1, \ldots, y_m]_\mathfrak{p}/\mathfrak{p}^n k[y_1, \ldots, y_m]_\mathfrak{p}[t_1, \ldots, t_d].$$

Hence $D'[t_1, \ldots, t_d]$ is a filtered colimit of smooth $R_\mathfrak{p}/\mathfrak{p}^n R_\mathfrak{p}$-algebras and $B \otimes_{k[x_1,\ldots,x_d]} (R_\mathfrak{p}/\mathfrak{p}^n R_\mathfrak{p})$ factors through one of these.

Ad (26). The final twist of the proof is that we cannot just use the map $B \to D'$ which maps x_i to the image of π_i' in D' and z_{ij} to the image of λ_{ij} in D' because we need the diagram

$$\begin{array}{ccc} B & \longrightarrow & D'[t_1, \ldots, t_d] \\ \uparrow & & \uparrow \\ k[x_1, \ldots, x_d] & \longrightarrow & R_\mathfrak{p}/\mathfrak{p}^n R_\mathfrak{p} \end{array}$$

to commute and we need the composition $B \to D'[t_1, \ldots, t_d] \to \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$ to be the map of (14). This requires us to map x_i to the image of $\pi_i t_i$ in $D'[t_1, \ldots, t_d]$. Hence we map z_{ij} to the image of $\lambda_{ij} t_i^{2N}/\delta_i^{2N}$ in $D'[t_1, \ldots, t_d]$ and everything is clear. $\square$